The Five Kingdoms of Life

Whittaker's Classification System — Biology Notes

Monera · Protista · Fungi · Plantae · Animalia

Introduction

In 1969, American ecologist **Robert Whittaker** proposed the Five Kingdom Classification system, grouping all living organisms into five kingdoms based on cell structure, mode of nutrition, body organization, and reproduction. This replaced the older two-kingdom system (Plant vs Animal), which failed to account for microorganisms and fungi properly.

Basis of Classification

- **Cell structure** — prokaryotic or eukaryotic
- **Body organization** — unicellular or multicellular; cellular, tissue, or organ level
- **Mode of nutrition** — autotrophic (self-feeding) or heterotrophic
- **Reproduction** — asexual or sexual
- **Phylogenetic relationships** — evolutionary relatedness among organisms

Overview Table

Kingdom	Cell Type	Nutrition	Organization
Monera	Prokaryotic	Autotrophic / Heterotrophic	Unicellular
Protista	Eukaryotic	Autotrophic / Heterotrophic	Unicellular
Fungi	Eukaryotic	Heterotrophic (saprophytic)	Multicellular (mostly)
Plantae	Eukaryotic	Autotrophic	Multicellular
Animalia	Eukaryotic	Heterotrophic	Multicellular

1. Kingdom Monera

Includes all **prokaryotic** organisms — cells lacking a true, membrane-bound nucleus and other membrane-bound organelles. This is the most primitive and abundant group of organisms on Earth.

Characteristics

- Unicellular, prokaryotic organisms
- No nuclear membrane; genetic material (DNA) lies free in the cytoplasm (nucleoid)
- Cell wall present, made of peptidoglycan (in bacteria)
- Reproduce mainly by binary fission; some by conjugation
- Nutrition varies — autotrophic (photosynthetic or chemosynthetic) or heterotrophic (saprophytic/parasitic)

Examples

- **Bacteria** — e.g. *Escherichia coli*, *Lactobacillus*
- **Cyanobacteria** (blue-green algae) — e.g. *Nostoc*, *Anabaena*

- **Archaebacteria** — extremophiles living in salty, acidic, or hot environments
- **Mycoplasma** — smallest living cells, lack a cell wall

2. Kingdom Protista

Consists of simple **eukaryotic** organisms, mostly unicellular. It acts as a link between the simplest monerans and more complex fungi, plants, and animals.

Characteristics

- Eukaryotic — true nucleus and membrane-bound organelles present
- Mostly unicellular; some form colonies
- Mode of nutrition varies widely — autotrophic, heterotrophic, or both
- Locomotion via cilia, flagella, or pseudopodia
- Reproduce both asexually (fission) and sexually

Sub-groups & Examples

Group	Examples	Feature
Protozoans (animal-like)	Amoeba, Paramecium	Heterotrophic, motile
Algae (plant-like)	Chlamydomonas, Diatoms	Photosynthetic
Slime moulds	Physarum	Saprophytic, produce spores

3. Kingdom Fungi

Eukaryotic, mostly multicellular organisms that lack chlorophyll and obtain nutrition by absorbing organic matter from their surroundings — either dead matter (saprophytic) or living hosts (parasitic).

Characteristics

- Cell wall made of **chitin** (not cellulose)
- Body made up of thread-like structures called **hyphae**, which together form a **mycelium**
- Heterotrophic — saprophytic, parasitic, or symbiotic (e.g. forming lichens with algae)
- Reproduce by spores — asexually (conidia, budding) and sexually
- Store food as glycogen, not starch

Examples

- **Yeast** — *Saccharomyces* (unicellular, used in baking & brewing)
- **Mould** — *Rhizopus* (bread mould), *Penicillium* (source of antibiotic penicillin)
- **Mushrooms** — *Agaricus*
- **Lichens** — symbiotic association of fungi and algae/cyanobacteria

4. Kingdom Plantae

Comprises all **multicellular, eukaryotic, autotrophic** organisms that manufacture their own food through photosynthesis using chlorophyll.

Characteristics

- Cell wall made of **cellulose**
- Contain chlorophyll and plastids; carry out photosynthesis
- Non-motile (fixed in one place)
- Store food as starch
- Show alternation of generations — alternating haploid (gametophyte) and diploid (sporophyte) phases

Major Divisions

Group	Key Feature	Example
Thallophyta (Algae)	No true root/stem/leaf	Spirogyra, Ulva
Bryophyta	'Amphibians of plant kingdom'; need water for fertilization	Moss, Marchantia
Pteridophyta	First plants with true roots, stem, leaves; no seeds	Fern, Horsetail
Gymnosperms	Naked seeds, not enclosed in fruit	Pine, Cycas
Angiosperms	Seeds enclosed in fruit; flowering plants	Mango, Wheat, Rose

5. Kingdom Animalia

Includes all **multicellular, eukaryotic, heterotrophic** organisms that lack cell walls and chlorophyll, and typically show locomotion and rapid response to stimuli.

Characteristics

- No cell wall; cells bounded only by a plasma membrane
- Heterotrophic nutrition — must consume other organisms for food
- Most show locomotion and well-developed sensory/nervous systems
- Reproduce mostly sexually; grow to a definite size and shape
- Body organization ranges from simple (cellular level) to complex (organ-system level)

Broad Classification

Group	Key Feature	Example
Porifera	Pore-bearing, no true tissues	Sponges

Cnidaria	Radial symmetry, stinging cells	Jellyfish, Hydra
Platyhelminthes	Flatworms, bilateral symmetry	Tapeworm, Planaria
Annelida	Segmented worms	Earthworm, Leech
Arthropoda	Jointed legs, exoskeleton — largest animal phylum	Insects, Spiders, Crabs
Mollusca	Soft body, often with a shell	Snail, Octopus
Echinodermata	Spiny skin, radial symmetry (adult)	Starfish, Sea urchin
Chordata	Notochord present at some stage; includes vertebrates	Fish, Frog, Birds, Humans

Quick Comparison Summary

Feature	Monera	Protista	Fungi	Plantae	Animalia
Cell type	Prokaryotic	Eukaryotic	Eukaryotic	Eukaryotic	Eukaryotic
Cell wall	Peptidoglycan	Varies	Chitin	Cellulose	Absent
Nutrition	Auto/Hetero	Auto/Hetero	Hetero	Auto	Hetero
Body form	Unicellular	Unicellular	Multicellular	Multicellular	Multicellular
Motility	Some motile	Motile	Non-motile	Non-motile	Mostly motile

Key Points to Remember

- The five kingdom system is based on **cell structure, nutrition, and body complexity**, proposed by **R.H. Whittaker (1969)**
- **Monera** is the only kingdom with prokaryotic cells
- **Fungi** cell walls contain chitin, unlike the cellulose walls of plants
- **Protista** serves as a bridge group with characteristics of both plants and animals
- Viruses are **not included** in any kingdom, as they are non-cellular and show properties of both living and non-living things